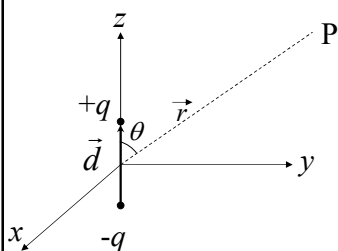
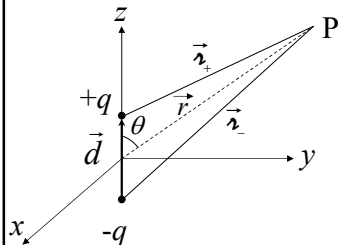


Simplest possible charge system



Potential ?
E. Field ?

Simplest possible charge system **Potential**



From cosine law:

$$r_+ = \left(r^2 + (d/2)^2 - rd \cos \theta \right)^{1/2}$$

$$\begin{aligned} r_- &= \left(r^2 + (d/2)^2 - rd \cos(\pi - \theta) \right)^{1/2} \\ &= \left(r^2 + (d/2)^2 + rd \cos \theta \right)^{1/2} \end{aligned}$$

$$r_{\pm} = \left(r^2 + (d/2)^2 \mp rd \cos \theta \right)^{1/2}$$

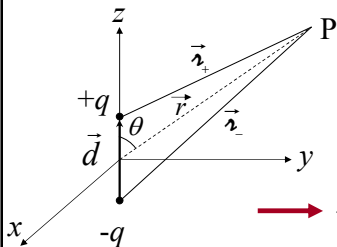
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

$$\frac{1}{r_{\pm}} = \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)^{-1/2}$$

$$\rightarrow \frac{1}{r_{\pm}} = \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \quad r \gg d$$

$$\approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

Simplest possible charge system **Potential**



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

$$\frac{1}{r_{\pm}} \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

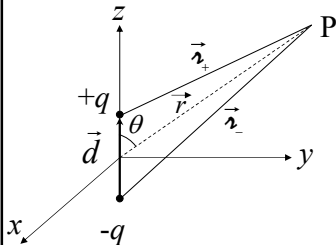
$$\begin{aligned} \rightarrow \frac{1}{r_+} - \frac{1}{r_-} &= \frac{1}{r} \left(1 + \frac{d}{2r} \cos \theta - 1 + \frac{d}{2r} \cos \theta \right) \\ &= \frac{d}{r^2} \cos \theta \end{aligned}$$

$$\rightarrow V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

But $d \cos \theta = \vec{d} \cdot \hat{r}$, $\vec{d} = d\hat{z}$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Simplest possible charge system **Potential**



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

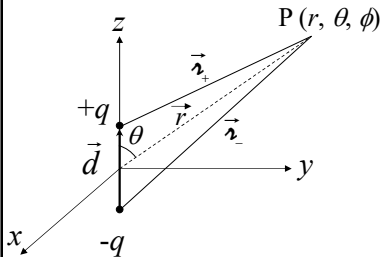
$$\vec{p} = q\vec{d} \quad \text{Dipole Moment}$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

or

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

Simplest possible charge system **Electric Field**



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2} \quad [1]$$

$$\vec{E} = -\vec{\nabla} V$$

$$\vec{E} = -\left[\hat{r} \frac{\partial V}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial V}{\partial \theta} \right] \quad [2]$$

Substituting (1) into (2), gives

$$\vec{E}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right] \quad [3]$$

or, in coordinate-free form

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \left[3(\vec{p} \cdot \hat{r}) \hat{r} - \vec{p} \right] \text{ HW}$$

Important Nomenclatures

Monopole



$$1/r^2$$

$$1/r$$

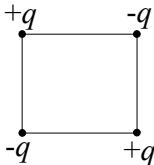
Dipole



$$1/r^3$$

$$1/r^2$$

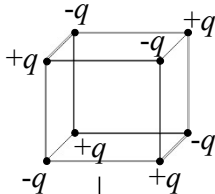
Quadrupole



$$1/r^4$$

$$1/r^3$$

Octopole



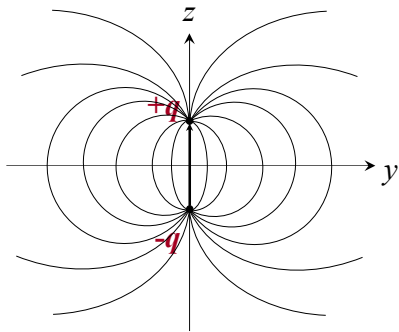
$$1/r^5$$

$$1/r^4$$

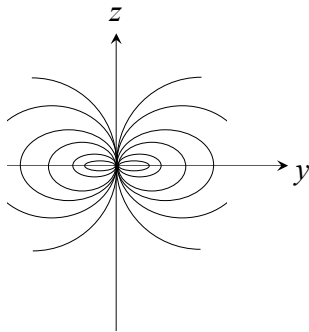
E. Field

Potential

Just to remind you:



Physical Dipole



Pure Dipole

**So far we have studied Electrostatics w. r. t.
individual charge/charge system/charge
distribution over a region and conductors**

**Study of Electrostatics is incomplete without
considering the effect of the matter – specially
how does matter respond towards an Electric field?**

- cause almost all applications of Electrostatics
involves field – matter interaction!**

Field – Matter Interaction

Matter can be solid (glass, metal, wood....), gas or liquid

- **Conductor:** having unlimited charges particles, not associated with the parent nucleus, and free to move
- **Insulator (DIELECTRICS):** every electron is attached to the atom/molecule.

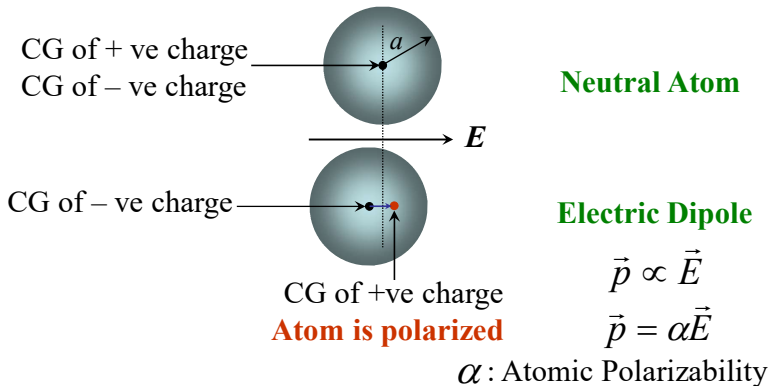
Can only get displaced within atom/molecule

Cumulative effect of this microscopic displacement becomes extremely imp

Cause, it defines the characteristic behavior of dielectric material towards an electric field

Field – Matter Interaction (Case I : Atom)

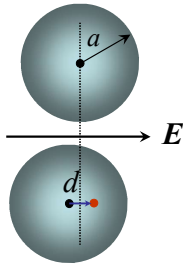
Question: What happens to a neutral atom when it is placed in an electric field?



Neutral Atom acquires tiny (induced) dipole moment $\vec{p}$

Field – Matter Interaction (Case I : Atom)

Question: What happens to a neutral atom when it is placed in an electric field?



Atom is polarized

Polarization is Unidirectional

In equilibrium

$$E_{\text{ext}} = E_{\text{electron cloud}}$$

$$|\vec{E}_{\text{electron cloud}}| = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}$$

$$|\vec{E}_{\text{ext}}| = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} = \frac{1}{4\pi\epsilon_0} \frac{p}{a^3}$$

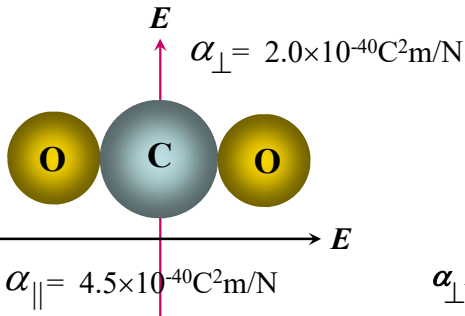
$$p = 4\pi\epsilon_0 a^3 E$$

$$p = 3\epsilon_0 (4\pi a^3 / 3) E$$

$$p = 3\epsilon_0 V E = \alpha E$$

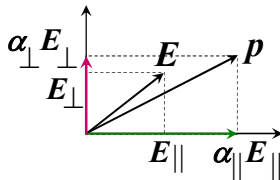
Field – Matter Interaction (Case II: Molecule)

Question: What happens to a linear molecule when it is placed in an electric field?



In general

$$p = \alpha_{\perp} E_{\perp} + \alpha_{\parallel} E_{\parallel}$$



Field – Matter Interaction (Case III: Molecule)

Question: What happens to a completely asymmetrical molecule when it is placed in an electric field?

$$\vec{p} = \alpha \vec{E}$$

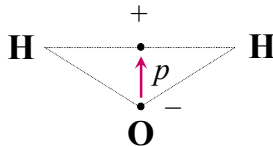
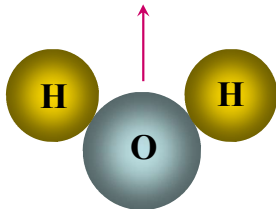
In general

$$\begin{aligned} p_x &= \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z \\ p_y &= \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z \\ p_z &= \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z \end{aligned} \quad \text{or} \quad \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

α_{ij} : Polarizability Tensor

Polarization is not linear

Field – Matter Interaction (Case IV: Polar Molecule)

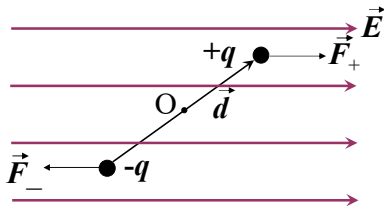


Polar Molecule

$$p = 6.1 \times 10^{-30} \text{C}\cdot\text{m}$$

Field – Matter Interaction (Case IV: Polar Molecule)

Question: polar molecule in uniform electric field



Net Force = 0

Net Torque (*about the center of the dipole*)

$$\begin{aligned}\vec{N} &= \vec{r}_+ \times \vec{F}_+ + \vec{r}_- \times \vec{F}_- \\ &= (\vec{d}/2) \times (q\vec{E}) + (-\vec{d}/2) \times (-q\vec{E}) = q\vec{d} \times \vec{E}\end{aligned}$$

$$\boxed{\vec{N} = \vec{p} \times \vec{E}}$$

Polarization:

Question: What happens to a dielectric material when it is placed in an electric field?

If made of neutral atoms/non-polar molecules, **field induces a tiny dipole moment** to each, in its own direction

If made of polar molecules, each permanent dipole will **experience torque** and align themselves in the direction of the field

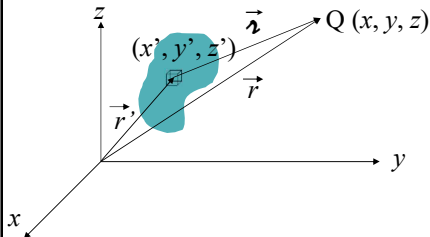


Lots of tiny dipoles pointing in the direction of the field

Material is polarized

dipole moment per unit volume = POLARIZATION ($\vec{P}$)

Field of a Polarized Dielectric



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{n}}{r^2}$$

$$\vec{p} = \vec{P}(\vec{r}') d\tau'$$

$$dV(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\hat{n} \cdot \vec{P}(\vec{r}')}{r^2} d\tau'$$

$$V(\vec{r}) = \int dV(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{n} \cdot \vec{P}(\vec{r}')}{r^2} d\tau'$$

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

$$\vec{\nabla}' \left(\frac{1}{r} \right) = \frac{\hat{n}}{r^2} \quad \Rightarrow \quad V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

Field of a Polarized Dielectric

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) = \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) + \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right)$$

$$\vec{P} \cdot \vec{\nabla}' \left(\frac{1}{r} \right) = \vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) - \frac{1}{r} (\vec{\nabla}' \cdot \vec{P})$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{\nabla}' \cdot \left(\frac{\vec{P}}{r} \right) d\tau' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\vec{P}}{r} \right) \cdot d\vec{a}' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

Field of a Polarized Dielectric

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\vec{P}}{r} \right) \cdot d\vec{a}' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau'$$

Potential due to
surface charge
distribution

Potential due to
volume charge
distribution

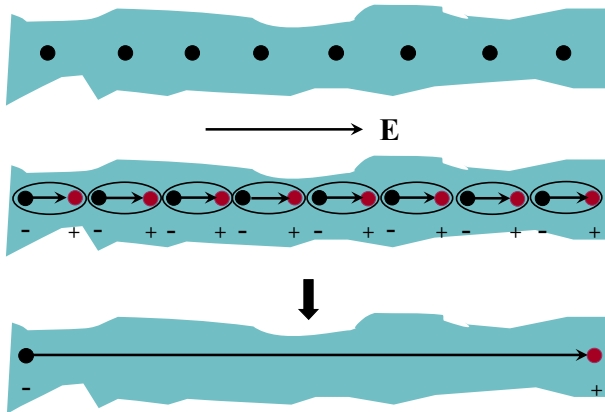
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \left(\frac{\sigma_b}{r} \right) da' + \int_V \frac{\rho_b}{r} d\tau'$$

→ $\sigma_b = \vec{P} \cdot \hat{n}$ **Bound surface charge density**

→ $\rho_b = -\vec{\nabla} \cdot \vec{P}$ **Bound volume charge density**

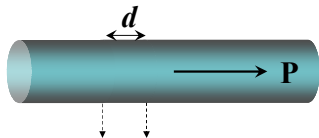
Physics behind the Bound Charge/Charge Density

Uniform Polarization



Physics behind the Bound Charge/Charge Density

Uniform Polarization



$$A \rightarrow = \bullet \text{---} \bullet$$

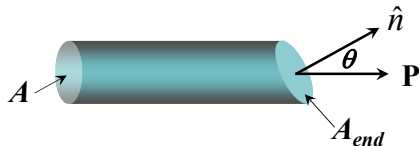
$-q \quad +q$

$$p = P(\cancel{Ad})$$

$$p = q\cancel{d}$$

$$q = PA$$

$$\frac{q}{A} = P = \sigma_b$$

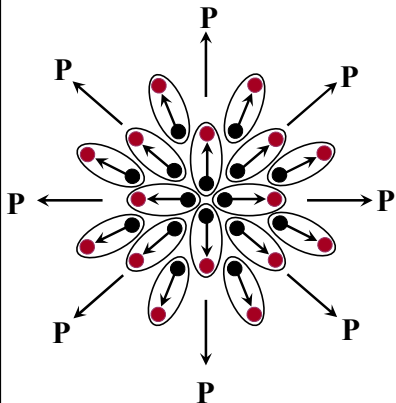


$$\sigma_b = \frac{q}{A_{end}} = P \cos \theta = \vec{P} \cdot \hat{n}$$

$$A = A_{end} \cos \theta$$

Physics behind the Bound Charge/Charge Density

Non-uniform Polarization



Divergence of $\mathbf{P}$ results in an accumulation of bound charge **within the material** and **on the surface**.

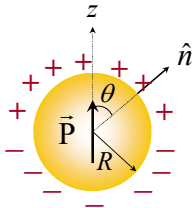
Essentially the net bound charge piling up throughout the volume *is equal and opposite* to the amount that has been pushed out from the surface.

$$\int_V \rho_b d\tau = -\oint_S \vec{\mathbf{P}} \cdot d\vec{\mathbf{a}} = -\int_V (\vec{\nabla} \cdot \vec{\mathbf{P}}) d\tau$$

→ $\rho_b = -\vec{\nabla} \cdot \vec{\mathbf{P}}$

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0 \quad \text{cause } \vec{P} \text{ is uniform}$$

$$\sigma_b = \vec{P} \cdot \hat{n} = P \cos \theta$$



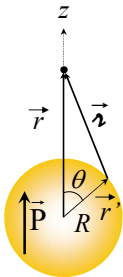
Field by σ_b plastered over the surface of a sphere

Borrowing the result from
BVP in spherical polar CS
(Ex. 3.9)

$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

Example

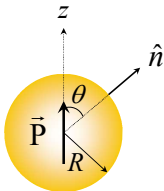
$\vec{E}$ produced by a uniformly polarized sphere of radius R
Homework



$$\begin{aligned} V(z) &= \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{P \cos\theta R^2 \sin\theta d\theta d\varphi}{(z^2 + R^2 - 2zR \cos\theta)^{1/2}} \\ &= \frac{PR^2}{2\epsilon_0} \int_{-1}^1 \frac{t dt}{(z^2 + R^2 - 2zRt)^{1/2}} \end{aligned}$$

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

$r < R$

$$r \cos \theta = z$$

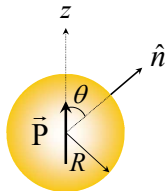
$$V = \frac{P}{3\epsilon_0} r \cos \theta = \frac{P}{3\epsilon_0} z$$

$$\longrightarrow \vec{E} = -\vec{\nabla} V = -\frac{P}{3\epsilon_0} \hat{z}$$

Field is constant inside

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & r \leq R \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r \geq R \end{cases}$$

$r > R$

$$V = \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta$$

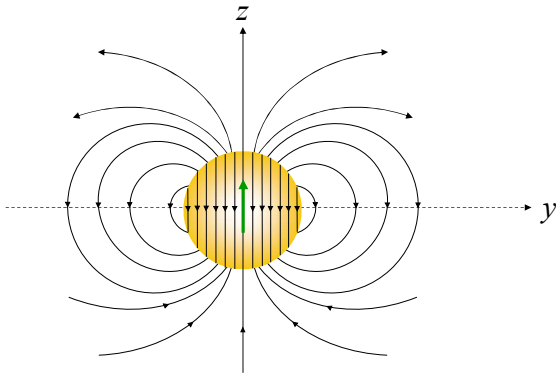
but $P = \frac{p}{(4\pi R^3)/3}$

$$\rightarrow V = \frac{p}{\cancel{3\epsilon_0} [4\pi \cancel{R^3} / \cancel{3}]} \frac{\cancel{R^3}}{r^2} \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

Potential of a perfect dipole at the origin!

Example

$\vec{E}$ produced by a uniformly polarized sphere of radius R



Dipole moment of the sphere $\vec{p} = \frac{4\pi R^3}{3} \vec{P}$

Gauss's Law in the Presence of Dielectric

Effect of Polarization is to produce accumulation of bound charges $\sigma_b = \vec{P} \cdot \hat{n}$ **on the surface of the dielectric**

$$\rho_b = -\vec{\nabla} \cdot \vec{P} \quad \text{within the dielectric}$$

Field due to polarization of the medium is just the field of these bound charges

Let's put together the cumulative effect of entire charges (bound + free charges)

Total charge density within the dielectric is $\rho = \rho_b + \rho_f$

$$\epsilon_0 \vec{\nabla} \cdot \vec{E} = \rho = -\vec{\nabla} \cdot \vec{P} + \rho_f \rightarrow \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f \quad \epsilon_0 \vec{E} + \vec{P} = \vec{D}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\int_V \vec{\nabla} \cdot \vec{D} d\tau = \int_V \rho_f d\tau$$

$$\int_S \vec{D} \cdot d\vec{a} = Q_{f \text{ enc}}$$

Susceptibility, Permittivity, Dielectric constant

$$\vec{P} \propto \vec{E}$$

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon \vec{E}$$

$$\vec{D} = \epsilon \vec{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

In vacuum, $\chi_e = 0 \rightarrow \epsilon = \epsilon_0$

In conductor, $\chi_e \rightarrow \infty$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = (1 + \chi_e)$$

Relative permittivity/Dielectric constt

Few Important points

$$\varepsilon \neq \varepsilon(\vec{r})$$

Homogeneous Medium

$$\varepsilon = \varepsilon(\vec{r})$$

Inhomogeneous Medium

$$\varepsilon \neq \varepsilon(\vec{r}, \vec{E}) \quad \vec{P} = \varepsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \varepsilon \vec{E}$$

Linear Medium

$$\varepsilon = \varepsilon(\vec{E}) \quad \vec{P} = \varepsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \varepsilon \vec{E}$$

Nonlinear Medium

$$\varepsilon = \text{Constant}$$

$$\vec{D} \parallel \vec{E} \quad \vec{P} \parallel \vec{E}$$

Isotropic Medium

$$\varepsilon = \text{Second rank tensor}$$

$$\vec{D} \nparallel \vec{E} \quad \vec{P} \nparallel \vec{E}$$

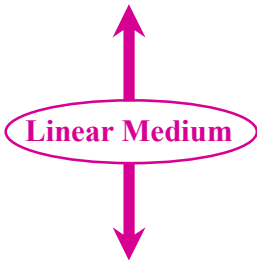
Anisotropic Medium

Few Important points

$$\epsilon \neq \epsilon(\vec{r})$$

Homogeneous Medium

$$\epsilon \neq \epsilon(\vec{r}, \vec{E}) \quad \vec{P} = \epsilon_0 \chi_e \vec{E}$$
$$\vec{D} = \epsilon \vec{E}$$



$$\epsilon = \text{Constant}$$

$$\vec{D} \parallel \vec{E} \quad \vec{P} \parallel \vec{E}$$

Isotropic Medium

Summary

With dielectric

$$\vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\vec{\nabla} \times \vec{E} = 0$$

which is the macroscopic counterpart of microscopic equations

Without dielectric

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = 0$$

as if we replaced $\rho \rightarrow \rho_f$ $\epsilon_0 \vec{E} \rightarrow \vec{D}$

$\vec{D}$ is just like $\vec{E}$ and we can calculate it using

~~$$\vec{D}(\vec{r}) = \frac{1}{4\pi} \int_V \frac{\hat{r}}{r^2} \rho_f(\vec{r}') d\tau'$$~~

because $\epsilon_0 \vec{E} + \vec{P} = \vec{D}$ and $\vec{\nabla} \cdot \vec{D} = \rho_f$ gives only the divergence

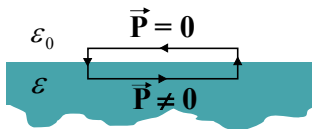
We need to find curl. $\vec{\nabla} \times \vec{D} = \epsilon_0 (\vec{\nabla} \times \vec{E}) + (\vec{\nabla} \times \vec{P}) = (\vec{\nabla} \times \vec{P}) \neq 0$

Question

Consider a Linear, Homogeneous, Isotropic Dielectric

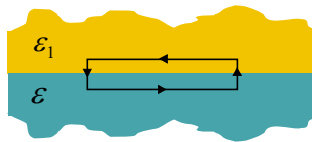
$$\vec{\mathbf{D}} = \epsilon_0 \vec{\mathbf{E}} + \vec{\mathbf{P}} \quad \vec{\mathbf{P}} = \epsilon_0 \chi_e \vec{\mathbf{E}}$$

$\vec{\mathbf{P}}$ and $\vec{\mathbf{D}} \propto \vec{\mathbf{E}}$ $\longrightarrow \vec{\nabla} \times \vec{\mathbf{D}} = \vec{\nabla} \times \vec{\mathbf{P}} = 0$?



$$\oint \vec{\mathbf{P}} \cdot d\vec{\mathbf{l}} \neq 0$$

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0$$



$$\oint \vec{\mathbf{P}} \cdot d\vec{\mathbf{l}} \neq 0$$

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0$$

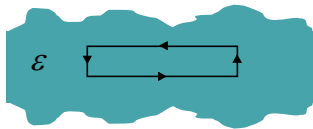
Question

Consider a Linear, Homogeneous, Isotropic Dielectric

$$\vec{\mathbf{D}} = \epsilon_0 \vec{\mathbf{E}} + \vec{\mathbf{P}} \quad \vec{\mathbf{P}} = \epsilon_0 \chi_e \vec{\mathbf{E}}$$

$\vec{\mathbf{P}}$ and $\vec{\mathbf{D}} \propto \vec{\mathbf{E}}$ $\longrightarrow \vec{\nabla} \times \vec{\mathbf{D}} = \vec{\nabla} \times \vec{\mathbf{P}} = 0$?

However, if the region is completely filled with a linear, homogeneous, isotropic dielectric

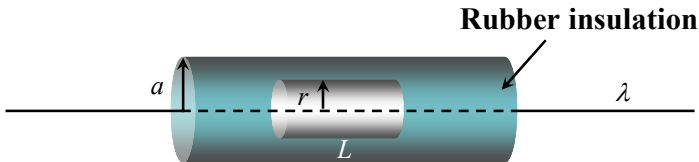


$$\oint \vec{\mathbf{P}} \cdot d\vec{\mathbf{l}} = 0$$

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0$$

$$\vec{\nabla} \times \vec{\mathbf{P}} = \vec{\nabla} \times \vec{\mathbf{D}} = 0 \longrightarrow \boxed{\vec{\nabla} \cdot \vec{\mathbf{D}} = \rho_f \quad \vec{\nabla} \times \vec{\mathbf{D}} = 0}$$

Example: 1



Find electric displacement.

$$\int_S \vec{D} \cdot d\vec{a} = Q_{f\,enc}$$

$$D(2\pi r L) = \lambda L$$

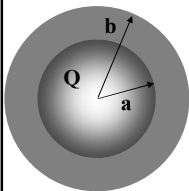
$$\vec{D} = \frac{\lambda}{2\pi r} \hat{r} \quad \text{valid inside/outside}$$

$$\text{Outside } \vec{P} = 0 \quad \epsilon_0 \vec{E} + \vec{P} = \vec{D} \rightarrow \epsilon_0 \vec{E} = \vec{D} \quad \vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

Inside, we do not know $\vec{P}$, and hence can not find $\vec{E}$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



To compute $V \longrightarrow$ need $\vec{E}$

To compute $\vec{E} \longrightarrow$ need bound charge

To compute bound charge $\longrightarrow$ need $\vec{P}$

To compute $\vec{P} \longrightarrow$ need $\vec{E}$?

What we have: *free* charge Q

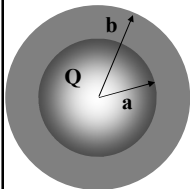
: spherical symmetry



Calculate $\vec{D}$

Example: 2

Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?



For $r > a$

$$\oint \vec{D} \cdot d\vec{a} = Q_{f \text{ enc}}$$

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r}$$

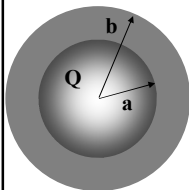
Inside the metal sphere, $\vec{E} = \vec{P} = \vec{D} = 0$

As $\vec{D} = \epsilon \vec{E}$

$$\vec{E} = \begin{cases} \frac{Q}{4\pi \epsilon r^2} \hat{r} & \text{for } a < r < b \\ \frac{Q}{4\pi \epsilon_0 r^2} \hat{r} & \text{for } r > b \end{cases}$$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**

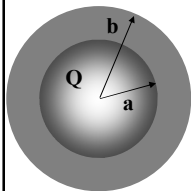


Potential at the center is

$$\begin{aligned} V &= -\int_{\infty}^0 \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} \\ &= -\int_{\infty}^b \left(\frac{Q}{4\pi\epsilon_0 r^2} \right) dr - \int_b^a \left(\frac{Q}{4\pi\epsilon r^2} \right) dr - \int_a^0 (0) dr \\ &= \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} + \frac{1}{\epsilon b} \right) \end{aligned}$$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Potential at the center?**



Potential at the center is

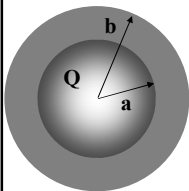
$$V = \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} + \frac{1}{\epsilon b} \right)$$

we got it without computing $\vec{P}$ or bound charges

Lets now compute $\vec{P}$ and the bound charges

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
 $\vec{P}$? Must essentially be defined in the dielectric region**



$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

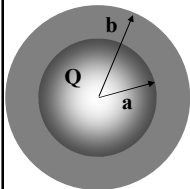
But
$$\vec{E} = \frac{Q}{4\pi\epsilon r^2} \hat{r} \quad \text{for } a < r < b$$

$$\text{Hence, } \vec{P} = \epsilon_0 \chi_e \vec{E} = \frac{\epsilon_0 \chi_e Q}{4\pi\epsilon r^2} \hat{r}$$

But how about bound charges ?

Example: 2

Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:



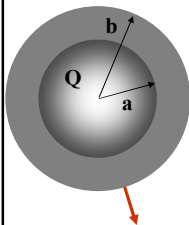
$$\vec{\mathbf{P}} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{\mathbf{P}} = 0 \quad \because \quad \vec{\nabla} \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 0$$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .**

Bound charge densities:



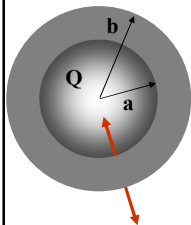
$$\vec{\mathbf{P}} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{\mathbf{P}} = 0$$

$$\sigma_b = \vec{\mathbf{P}} \cdot \hat{\mathbf{n}} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon b^2} & \text{at the outer surface} \end{cases}$$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



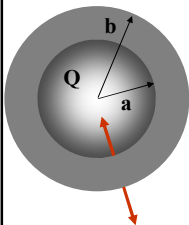
$$\vec{\mathbf{P}} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{\mathbf{P}} = 0$$

$$\sigma_b = \vec{\mathbf{P}} \cdot \hat{\mathbf{n}} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon b^2} & \text{at the outer surface} \end{cases}$$

Example: 2

**Metal sphere of radius a carries a charge Q .
is surrounded out to a radius b , by a linear dielectric ϵ .
Bound charge densities:**



$$\vec{\mathbf{P}} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{\mathbf{P}} = 0$$

$$\sigma_b = \vec{\mathbf{P}} \cdot \hat{\mathbf{n}} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon b^2} & \text{at the outer surface} \\ -\frac{\epsilon_0 \chi_e Q}{4\pi \epsilon a^2} & \text{at the inner surface} \end{cases}$$

Boundary Conditions

Recall the main relations $\vec{\nabla} \cdot \vec{\mathbf{E}} = \rho / \varepsilon_0$ $\vec{\nabla} \times \vec{\mathbf{E}} = \mathbf{0}$

These relations give rise to the boundary conditions:

$$\mathbf{E}_{\text{above}}^{\perp} - \mathbf{E}_{\text{below}}^{\perp} = \sigma / \varepsilon_0$$

$$\mathbf{E}_{\text{above}}^{\parallel} = \mathbf{E}_{\text{below}}^{\parallel}$$

$$V_{\text{above}} = V_{\text{below}}$$

Boundary Conditions in Dielectrics

$$\vec{\nabla} \cdot \vec{\mathbf{D}} = \rho_f \quad \longrightarrow \quad D^\perp_{\text{above}} - D^\perp_{\text{below}} = \sigma_f$$

$\longrightarrow$ **$D^\perp$ is discontinuous across the boundary**

Or more conveniently: $\epsilon_{\text{above}} E^\perp_{\text{above}} - \epsilon_{\text{below}} E^\perp_{\text{below}} = \sigma_f$

$$\epsilon_{\text{above}} \frac{\partial V_{\text{above}}}{\partial n} - \epsilon_{\text{below}} \frac{\partial V_{\text{below}}}{\partial n} = -\sigma_f \quad \textbf{E}^\perp \text{ is discontinuous across the boundary}$$

$$\oint \vec{\mathbf{E}} \cdot d\vec{l} = 0 \quad \textbf{E}^\parallel \text{ is always continuous across the boundary}$$

$$\vec{\nabla} \times \vec{\mathbf{D}} = \vec{\nabla} \times \vec{\mathbf{P}} \quad \longrightarrow \quad D^\parallel_{\text{above}} - D^\parallel_{\text{below}} = P^\parallel_{\text{above}} - P^\parallel_{\text{below}}$$

$D^\parallel$ is discontinuous across the boundary

$$V_{\text{above}} = V_{\text{below}} \quad \textbf{Potential is continuous across the boundary}$$

Boundary Conditions

In the absence of any free charge

$$\vec{\nabla} \cdot \vec{\mathbf{D}} = 0 \quad \longrightarrow \quad D_{\text{above}}^{\perp} - D_{\text{below}}^{\perp} = 0$$

$$\epsilon_{\text{above}} E_{\text{above}}^{\perp} - \epsilon_{\text{above}} E_{\text{below}}^{\perp} = 0$$

$$\epsilon_{\text{above}} \frac{\partial V_{\text{above}}}{\partial n} - \epsilon_{\text{above}} \frac{\partial V_{\text{below}}}{\partial n} = 0$$

And always $V_{\text{above}} = V_{\text{below}}$

$D^{\perp}$ is always continuous across the boundary

$E^{\perp}$ is discontinuous across the boundary

$\mathbf{D}^{\parallel}$ is discontinuous across the boundary

$\mathbf{E}^{\parallel}$ is always continuous across the boundary

Question

Consider a Linear, Homogeneous, Isotropic Dielectric

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$\vec{P}$ and $\vec{D} \propto \vec{E}$ $\longrightarrow \vec{\nabla} \times \vec{D} = \vec{\nabla} \times \vec{P} = 0$?

However, if the region is completely filled with a linear, homogeneous, isotropic dielectric

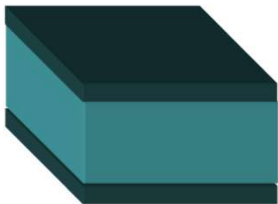
$$\vec{D} = \epsilon \vec{E} \quad \epsilon = \epsilon_0 (1 + \chi_e) \quad \vec{\nabla} \cdot \vec{D} = \rho_f \rightarrow \vec{\nabla} \cdot (\epsilon \vec{E}) = \rho_f$$

$$\rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho_f}{\epsilon} \rightarrow \oint \vec{E} \cdot d\vec{a} = \int \frac{\rho_f}{\epsilon} d\tau \rightarrow E 4\pi r^2 = \frac{1}{\epsilon} \int_V \rho_f d\tau$$

$$\rightarrow E = \frac{1}{4\pi\epsilon r^2} \int_V \rho_f d\tau \rightarrow E = \frac{\epsilon_0}{\epsilon} \frac{1}{4\pi\epsilon_0 r^2} \int_V \rho_f d\tau \rightarrow \vec{E} = \frac{\epsilon_0}{\epsilon} \vec{E}_{\text{vac}}$$

$$\vec{D} = \epsilon_0 \vec{E}_{\text{vac}}$$

Example: 3 Parallel Plate Capacitor



Dielectric

$$C = Q/V$$

$$\vec{\mathbf{E}} = \frac{1}{\epsilon_r} \vec{\mathbf{E}}_{\text{vac}} \quad \rightarrow \quad V = \frac{1}{\epsilon_r} V_{\text{vac}}$$

$$C = \epsilon_r C_{\text{vac}}$$

Energy in Dielectric Systems

Capacitor $C = \frac{Q}{V}$

Work to charge up a Capacitor $W = \frac{1}{2}CV^2$

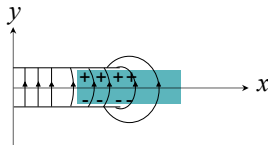
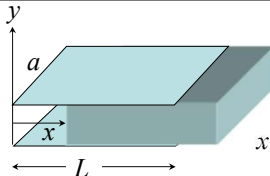
If the capacitor is filled with a linear dielectric $C = \epsilon_r C_{\text{vac}}$

→ **Work needed to charge up a Capacitor also increases by ϵ_r**

Recall $W = \frac{\epsilon_0}{2} \int E^2 d\tau$

$$W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau = \frac{\epsilon_0}{2} \int \frac{\epsilon}{\epsilon_0} E^2 d\tau = \frac{1}{2} \int \epsilon E^2 d\tau = \frac{1}{2} \int \vec{D} \cdot \vec{E} d\tau$$

Forces on Dielectrics



Assuming total charge on the plate to be constt

$$dW = F_{ext} dx$$

$$F_{ext} = -F$$

$$F = -\frac{dW}{dx}$$

$$W = \frac{1}{2} CV^2$$

$$C = \frac{\epsilon_0 a}{d} (\epsilon_r l - \chi_e x)$$

$$W = \frac{1}{2} \frac{Q^2}{C}$$

$$F = -\frac{dW}{dx} = -\frac{1}{2} \frac{Q^2}{C^2} \frac{dC}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}$$

$$\frac{dC}{dx} = -\frac{\epsilon_0 \chi_e a}{d}$$

$$F = -\frac{\epsilon_0 \chi_e a}{2d} V^2$$

